



NAAN MUDHALVAN ARTS INTERNSHIP PROGRAM 2026

Frontend Web Development using React

Project Submission Template

STUDENT NAME: P.Janani

YEAR: III rd Year

COLLEGE CODE: 303

COLLEGE NAME: Arcot Sri Mahalakshmi Women's College

DEPARTMENT: BCA

SEMESTER: 5

ROLL NO: 30324U09028

NMID: 2715C4DA34F37E98D0E4B8DE8F06B59D

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TITLE: HR Employee Leave Management tool

GITHUB LINK : <https://github.com/jananipachaiyappan3-eng/Janani..p.git>

1.ABSTRACT

Abstract – HR Employee Leave Management Tool

The HR Employee Leave Management Tool is a web-based application designed to simplify and automate the process of managing employee leave requests within an organization. The system allows employees to submit leave applications online, view their leave status, and track available leave balances. HR administrators and managers can review, approve, reject, and maintain leave records efficiently through a centralized platform.

The tool reduces paperwork, minimizes manual errors, and improves transparency in leave management. It provides real-time updates on leave requests and maintains a digital database of employee leave information for easy access and reporting. The system is developed using web technologies such as HTML, CSS, and JavaScript, offering a user-friendly interface and smooth functionality.

Overall, the HR Employee Leave Management Tool enhances organizational productivity by streamlining leave administration, ensuring accurate record keeping, and improving communication between employees and management. It serves as an efficient solution for modern HR operations and workforce management.

2.PROBLEM STATEMENT

Problem Statement – HR Employee Leave Management Tool

In many organizations, employee leave management is still handled manually through paper forms, emails, or spreadsheets. This traditional approach is time-consuming, prone to errors, and makes it difficult to track leave balances, approvals, and employee attendance accurately. Employees often face delays in receiving approval updates, while HR departments struggle to maintain organized records and generate reports efficiently.

The lack of a centralized leave management system can lead to miscommunication, inaccurate leave calculations, data redundancy, and reduced productivity. Therefore, there is a need for an automated solution that simplifies the leave application and approval process, maintains accurate leave records, provides real-time status updates, and improves overall efficiency in human resource management.

3.OBJECTIVES

The main objective of the HR Employee Leave Management Tool is to automate and simplify the employee leave process.

It enables employees to apply for leave online, allows managers to approve or reject requests efficiently, and maintains accurate leave records in a centralized system. The tool reduces paperwork, saves time, improves transparency, and helps organizations manage employee attendance and leave policies effectively.

4.PROJECT SCOPE

Project Scope of HR Employee Leave Management Tool

The HR Employee Leave Management Tool is designed to automate and streamline the leave management process within an organization. The system allows employees to apply for various types of leave, view their leave balance, and track the status of their requests. Managers and HR administrators can review, approve, reject, and monitor leave applications through a centralized platform.

The project scope also includes maintaining employee leave records, generating leave reports, managing leave policies, and providing notifications for leave requests and approvals. The system aims to reduce manual paperwork, improve accuracy, enhance transparency, and ensure efficient management of employee attendance and leave information across the organization.

5. TECHNOLOGIES USED

Technologies Used in HR Employee Leave Management Tool

The HR Employee Leave Management Tool is developed using the following technologies:

HTML (HyperText Markup Language) – Used to create the structure and layout of the web pages.

CSS (Cascading Style Sheets) – Used for designing and styling the user interface.

JavaScript – Used to add interactivity and validate user inputs.

PHP – Used as the server-side scripting language to process leave requests and manage system operations.

MySQL – Used as the database to store employee details, leave records, and approval information.

XAMPP – Used as a local server environment for developing and testing the application.

These technologies work together to provide a user-friendly, secure, and efficient leave management system for employees and HR administrators.

6. APPLICATION TOOL

The HR Employee Leave Management Tool is a software application designed to automate and simplify the leave management process within an organization. It enables employees to apply for leave online, check their leave balances, and track the status of their requests. Managers and HR personnel can review, approve, or reject leave applications efficiently through a centralized system. The tool maintains accurate records of employee leave history, generates reports, and reduces paperwork by replacing manual processes. By ensuring compliance with company leave policies and improving communication between employees and management, the system enhances productivity, accuracy, and overall workforce management.

7.KEY REACT CONCEPTS USED

The HR Employee Leave Management Tool utilizes several key React concepts to create an efficient and interactive user interface. React Components are used to build reusable sections such as the login page, employee dashboard, leave application form, and leave history page. JSX (JavaScript XML) helps in designing the user interface with a simple and readable syntax. The useState hook is used to manage dynamic data such as employee details, leave requests, and form inputs, while the useEffect hook handles data fetching and updates from the backend. Props are used to pass information between components, enabling smooth communication within the application. Event handling manages user actions such as submitting leave requests and approving applications. Conditional rendering displays different content based on user roles, such as Employee, Manager, or HR Administrator. React Router facilitates seamless navigation between different pages of the application, and API integration allows the frontend to communicate with the backend database for storing and retrieving leave management data. Together, these React concepts ensure a responsive, scalable, and user-friendly leave management system.

8.EXPECTED OUTPUT

Expected Output of HR Employee Leave Management Tool

The HR Employee Leave Management Tool provides a simple and efficient system for managing employee leave requests. Employees can log in, apply for leave by selecting leave dates and leave type, and track the status of their requests. The system displays leave balances, pending requests, approved leaves, and rejected applications in an organized manner.

For HR administrators, the tool allows viewing all employee leave requests, approving or rejecting applications, maintaining leave records, and generating reports. The expected output is a user-friendly platform that reduces paperwork, improves accuracy in leave tracking, ensures transparency in the approval process, and enhances overall workforce management efficiency.

9. CHALLENGES FACED

Challenges Faced by HR Employee Leave Management Tool

The development of the HR Employee Leave Management Tool faced several challenges. One of the major challenges was maintaining accurate leave records and ensuring that leave balances were updated correctly after every leave request, approval, or cancellation. Handling different types of leave such as casual leave, sick leave, and earned leave also required proper validation and management.

Another challenge was implementing a user-friendly interface that allows employees and HR administrators to easily submit, review, and approve leave requests. Ensuring data security, preventing unauthorized access, and maintaining reliable database connectivity were also important concerns. Additionally, generating reports, managing multiple users simultaneously, and reducing manual errors in the leave approval process required careful planning and efficient system design.

10. LEARNING OUTCOMES

Learning Outcomes of HR Employee Leave Management Tool

The development of the HR Employee Leave Management Tool provided valuable learning experiences in both software development and human resource management processes. Through this project, knowledge was gained in designing and developing a web-based application using HTML, CSS, JavaScript, and database technologies. It also helped in understanding concepts such as user authentication, form validation, data management, and report generation.

Additionally, the project enhanced problem-solving, debugging, and teamwork skills while implementing leave request and approval workflows. It provided practical experience in database operations, data security, and creating user-friendly interfaces. The project also improved understanding of how technology can automate HR activities, reduce manual work, increase accuracy, and improve overall organizational efficiency.

Github link:

<https://github.com/jananipachaiyappan3-eng/Janani..p.git>

11.CONCLUSION

Conclusion of HR Employee Leave Management Tool

The HR Employee Leave Management Tool is a useful and efficient application that automates the process of managing employee leave requests within an organization. The system eliminates the need for manual paperwork and helps maintain accurate leave records in a centralized database. Employees can easily apply for leave, check their leave status, and view their leave balance, while HR administrators can review, approve, or reject requests efficiently. This project demonstrates how modern web technologies can be used to simplify HR operations and improve workplace productivity. By reducing manual errors, saving time, and ensuring transparency in the leave approval process, the system enhances overall organizational efficiency. The project also provides a scalable foundation that can be extended with additional HR features in the future. Overall, the HR Employee Leave Management Tool serves as a reliable solution for effective leave management and contributes to a more organized and digitally managed work environment.